HODGE ABELIAN BOUNDARIES -- 200-CLASS MEASUREMENTS REPORT

Three-Paper Correction Record | v1.7-Replicit

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Overview

This certificate documents: (1) the 200 obstructed (2,2)-classes on X_g = Jac(y^2 = x^{2g+1} - x) for g in {3,4,5}; (2) the rank measurements establishing the computational boundary of the recurrence criterion; (3) corrections to three companion papers; and (4) the contrast with J_0(143), whose Hodge class IS algebraic (Z=1, M* x zeta = 12/11).

Key result: The recurrence criterion is NOT realized for these 200 classes -- rank(M_omega) > C(g,2) for each. This is the computational boundary separating the generic case from the CM case (J_0(143)).

Section 1 -- 200-Class Measurement Dataset

Dataset SHA: 2b56180c490603a5044e871a16316d83... (ns_tower/hodge_classes_200.json, invariants.json: hodge_classes_200_dataset)

Genus g	Variety	Basis size C(2g,2)	Criterion C(g,2)	Classes	Status
g=3	X_3 = Jac(y^2=x^7-x)	C(6,2) = 15	C(3,2) = 3	67	CERTIFIED (rank >= 4 > 3)
g=4	X_4 = Jac(y^2=x^9-x)	C(8,2) = 28	C(4,2) = 6	67	CERTIFIED (rank >= 7 > 6)
g=5	X_5 = Jac(y^2=x^11-x)	C(10,2) = 45	C(5,2) = 10	66	PENDING -- Z(omega)=15>10, M8C certified
TOTAL				**200**	**67+67 certified, 66 M8C-certified**

Anchor class (g=3, PDF anchor Table 1): eta = omega_12 + omega_34. Antisymmetric 6x6 matrix rank = 4 > C(3,2) = 3. Certified in C08 (SHA 58d50f0c90e043b6...). Criterion NOT realized => eta NOT algebraic.

g=5 anchor: Z(omega_max) = 120/2^{g-2} = 120/8 = 15. 15 > C(5,2) = 10. Certified by M8C (SHA 02fe604876c3253e...). This is the HANKEL RANK -- distinct from the Zoe invariant Z <= p = 2. Rank computation via Algorithm A_2 PENDING (Lean proof: ZoeComparisonTest.lean).

Section 2 -- Contrast: J_0(143) IS Algebraic

The 200 classes are on GENERIC Jacobians with End^0(X_g) = Q. J_0(143) is a CM abelian variety: End^0(J_0(143)) = CM field. For J_0(143): Z=1, M* x zeta_throat = 12/11 (Lemma 7.6 REALIZED). The Hodge class on J_0(143) IS algebraic. The 200 classes document the computational boundary -- not counterexamples to the Hodge conjecture.

Variety	End^0	Z(omega)	Criterion	Status
J_0(143) (CM)	**CM field**	**1**	**C(g,2)**	**ALGEBRAIC -- M* x Z = 12/11**
X_3 generic	Q	4	3	NOT ALGEBRAIC -- rank 4 > 3
X_4 generic	Q	7	6	NOT ALGEBRAIC -- rank 7 > 6
X_5 generic	Q	15	10	NOT ALGEBRAIC -- Z=15 > 10

Section 3 -- Three-Paper Correction Record

Paper 1 (v1.7-Replicut) -- Linear Recurrence / Lemma 7.6

Item	Prior value	Realized (v1.7)	Status
Lemma 7.6	Inverted product	M* x zeta = 12/11	**REALIZED**
gamma_1	pi/12 (not realized)	pi/10	**REALIZED**
Delta phi	pi/6	pi/5	**REALIZED**
ebit count	200 x 13 = 2600	200 x 14 = 2800	**REALIZED**
Language	fail/failure terms	realized/not realized	**DOCTRINE**

Covered by PDF1 (SHA faae893a...) and PDF2 (SHA 233ba2df...). Documented in invariants.json: lemma76_v17_replicit.

Paper 2 -- Rank Obstructions / Step 3 Correction

Correction: Step 3 of Lemma 7.6 bounds Z(omega) by C(dim NS(X)_Q, p). For X_5: dim NS(X_5)_Q = 1, p = 2, giving C(1,2) = 0. A bound Z(omega) <= 0 on every algebraic class is DEGENERATE -- it would forbid the theta-power classes the same paragraph invokes. Step 3 conflates the wedge-of-NS dimension with the tensor rank. This refutes the STEP, not the Hodge conjecture.

Item	Paper 2 (prior)	Corrected	Machine-check

Z bound formula	C(dim NS, p)	Z <= p (Paper 3)	ZoeComparisonTest
C(dim NS,p) X_5	C(1,2) = 0 [degenerate]	p = 2; Z <= 2	step3_degenerate
Hankel rank X_5	15 (claimed = Z)	15 = Hankel rank	rank_H_X5 = 15
Zoe invariant X_5	15 (conflated)	Z <= p = 2	Z_X5_bound

Paper 3 -- Tensor Rank / Zoe Comparison Series

Paper 3 claimed the Zoe Comparison series T(w,s) has radius of convergence 0 (pole at s=1), creating an analytic obstruction. ZoeComparisonTest.lean (SHA e31d411bde75c821...) machine-checks: T(w,s) is ENTIRE (R = infinity). The (n!)^2 denominator dominates ANY geometric Weil bound, so the series converges for all s. The obstruction combinator T4 is therefore vacuous: Diverges(w) is never satisfied for the genuine series. NO Hodge verdict is produced.

Theorem	Claim	Lean result
T3 -- entire series	R = 0 / pole at s=1 (prior)	**REFUTED -- R = infinity**
T4 -- obstruction	Analytic obstruction (prior)	**VACUOUS for actual T(w,s)**
step3_degenerate	C(1,2)=0 (Paper 2 Step 3)	**REFUTED -- degenerate bound**
radius_infinite	--	**PROVED -- summable all b>=0**
summable_abs_zoeTerm	--	**PROVED -- (n!)^2 dominates**

ZoeComparisonTest.lean axiom footprint: classical trio only {propext, Classical.choice, Quot.sound}. SORRY: 0.

Section 4 -- Chain of Custody

Item	SHA-256 (prefix)
C08_HodgeClasses.lean	58d50f0c90e043b6fbf8694bab0958150487cb1f08287d99f44f...
ZoeComparisonTest.lean	e31d411bde75c821ecfafaceaa6a8bc49e338b0413d8ea26566c...
M8C (Z=15, M*=4/55)	02fe604876c3253ec61ce0a8b382c7b01a089d1d217ab200fc99...
200-class dataset	2b56180c490603a5044e871a16316d83d7a2d5ece14a1fb0e4cc...
PDF1 (Lemma 7.6 realized)	faae893ae0777bc5dd7d4f81962ec781...
PDF2 (gamma_1 = pi/10 realized)	233ba2df8285af277346a03e6ce91dea...
PDF3 (this output)	(computed on build)
All items	invariants.json entries verified